

DATA MODELING USING POWER BI



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Introduction

This project focuses on transforming raw business data into meaningful insights using Microsoft Power BI. The datasets include *List of Orders*, *Order Details*, and *Sales Target*, which together provide a comprehensive view of sales transactions and performance targets.

Objectives of the Project

- To clean and prepare the datasets (*List of Orders*, *Order Details*, *Sales Target*) using Power Query.
- To calculate totals such as **Total Amount** and **Total Target by Month**.
- To build a **data model in Power BI** by creating relationships between tables.
- To generate clear insights that support decision-making.

Data Import & Transformation

- Connected to the source files containing the three datasets.
- Loaded the tables into Power Query for cleaning and shaping.

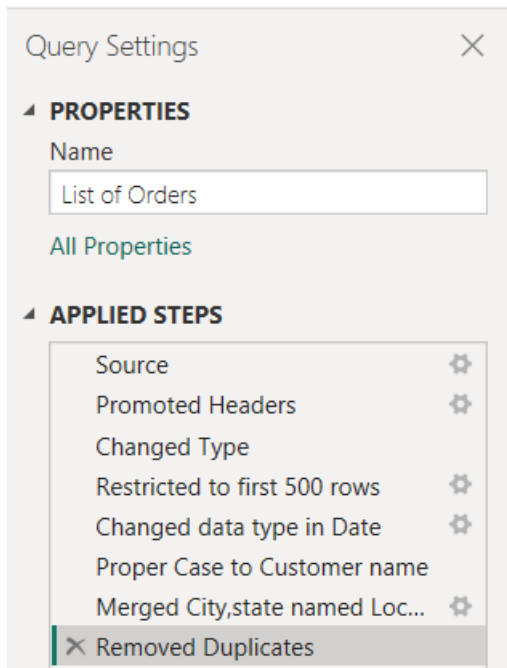
Transformation Logic

Transformation ensures the raw data is clean, consistent, and ready for analysis. In Power Query, the following steps are typically applied:

- **Data Cleaning** → Removing duplicates, handling missing values, and correcting data types.
- **Text Formatting** → Standardizing product names, categories, and labels for consistency.

Row Limiting & Data Type

- Restricted First 500 Rows



- **Changed data types** (e.g., Order Date -Date Type, Amount → Fixed Decimal Number, Target → Currency)

Order Date	
Valid	100%
Error	0%
Empty	0%
4/1/2018	
4/1/2018	
4/3/2018	
4/3/2018	
4/5/2018	

- Amount → Fixed Decimal Number

Data Type: Fixed decimal number	
Detect Data Type	Fill
Rename	Pivot Column
Any Column	
= Table.TransformColumns("#Trimmed 1	
Order ID	\$ Amount
Valid	100%
Error	0%
Empty	0%
-25601	1,275.00
-25602	168.00
-25603	1,355.00
-25604	65.00

- Target → Currency Data Type

Month of Order Date	Category	Target
4/18/2026	Furniture	10,400.00
5/18/2026	Furniture	10,500.00
6/18/2026	Furniture	10,600.00

• Text Formatting

- Consistency:** Ensuring all text values follow the same style.
- Readability:** Making names, categories, or labels clear and professional.
- Formatting the Customer name column to Proper Case to ensure consistent capitalization.

Query Settings	
PROPERTIES	
Name	List of Orders
All Properties	
APPLIED STEPS	
Source	⚙️
Promoted Headers	⚙️
Changed Type	
Restricted to first 500 rows	⚙️
Changed data type in Date	⚙️
Proper Case to Customer name	
Merged City,state named Loc...	⚙️
Removed Duplicates	✕

Merge Column

- **Purpose:** Combine two or more existing columns into a single column. Merged the City and State Column to a new column named (Location).

Location	
Valid	100%
Error	0%
Empty	0%
Gujarat,Ahmedabad	
Maharashtra,Pune	
Madhya Pradesh,Bhopal	
Rajasthan,Jaipur	
West Bengal,Kolkata	
Karnataka,Bangalore	
Jammu and Kashmir,Kashmir	
Tamil Nadu,Chennai	
Uttar Pradesh,Lucknow	
Bihar,Patna	

- **Custom Column**

Purpose: Create a new column using formulas or calculations. Created a custom column named as Profit margin calculated as profit / Amount=%.

% Profit Margin
-90.04%
-18.18%
-25.00%
-70.00%
-66.07%
-64.15%
43.98%
37.79%
-4.20%
-4.43%
-125.00%
-86.01%

- **Conditional Column**

A **Conditional Column** is used to create new values in a dataset based on specific rules or conditions. It works like an **IF...THEN...ELSE** statement in Excel or programming languages.



conditional column named Profit Status based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

Profit Status
Loss
Loss
Loss
Loss
Loss
Loss
Profit
Profit

Merging Data (Joins)

Merging data means combining two tables based on a common column (called a **key**). In Power Query, this is similar to performing a **JOIN** in databases. It allows you to bring related information together into one table for analysis.

Merged the List of Orders and Order Details tables using Order ID. The merged output table as Orders Data.

Queries [4] = Table.NestedJoin("#List of Orders", {"Order ID"}, "#Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)

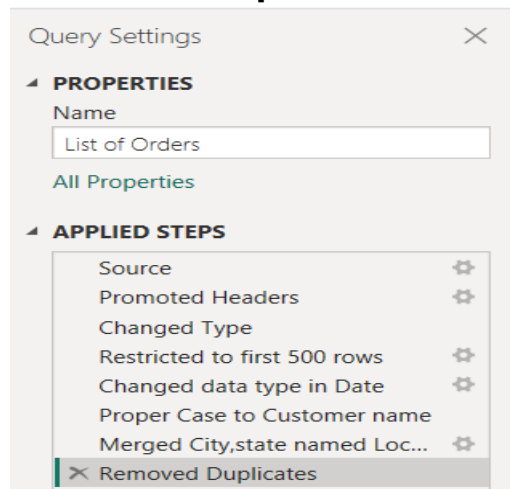
Order ID	Order Date	CustomerName	Location	Order Details
1 B-25601	4/1/2018	Bharat	Gujarat,Ahmedabad	Table
2 B-25602	4/1/2018	Pearl	Maharashtra,Pune	Table
3 B-25603	4/3/2018	Jahan	Madhya Pradesh,Bhopal	Table
4 B-25604	4/3/2018	Divsha	Rajasthan,Jaipur	Table
5 B-25605	4/5/2018	Kasheen	West Bengal,Kolkata	Table
6 B-25606	4/6/2018	Hazel	Karnataka,Bangalore	Table
7 B-25607	4/6/2018	Sonakshi	Jammu and Kashmir,Kashmir	Table
8 B-25608	4/8/2018	Aarushi	Tamil Nadu,Chennai	Table
9 B-25609	4/9/2018	Jitesh	Uttar Pradesh,Lucknow	Table
10 B-25610	4/9/2018	Yogesh	Bihar,Patna	Table
11 B-25611	4/11/2018	Anita	Kerala,Thiruvananthapuram	Table
12 B-25612	4/12/2018	Shrichand	Punjab,Chandigarh	Table
13 B-25613	4/12/2018	Mukesh	Haryana,Chandigarh	Table
14 B-25614	4/13/2018	Vandana	Himachal Pradesh,Simla	Table

Handling Missing Data & Duplicate Data

Strategies for handling remove duplicates → If they are errors.

- **Keep latest record** → If duplicates exist but only the most recent matters.
- **Aggregate values** → If duplicates represent multiple transactions, sum up *Quantity* or *Amount*.
- **Flag duplicates** → Mark them for review instead of deleting.

Removed Duplicates



Removed duplicates in Sales target

The screenshot shows the Power BI interface with a table of 12 rows. The columns are 'Month of Order Date', 'Category', and '\$ Target'. The 'Sales target' column is highlighted, and the 'Removed Duplicates' step is applied to it. The table data is as follows:

	Month of Order Date	Category	\$ Target
1	26/04/18	Furniture	10,400.00
2	26/05/18	Furniture	10,500.00
3	26/06/18	Furniture	10,600.00
4	26/07/18	Furniture	10,800.00
5	26/08/18	Furniture	10,900.00
6	26/09/18	Furniture	11,000.00
7	26/10/18	Furniture	11,100.00
8	26/11/18	Furniture	11,300.00
9	26/12/18	Furniture	11,400.00
10	26/01/19	Furniture	11,500.00
11	26/02/19	Furniture	11,600.00
12	26/03/19	Furniture	11,800.00

Sorting Data

- **Purpose:** Arrange rows in a logical order for easier analysis.
- Select the column to filter (e.g., *State*, *Category*, *Profit*).
- A filter options (equals, contains, greater than, less than, etc.).

Outcome

- **Sorting** organizes data for better readability.
- **Filtering** narrows down the dataset to focus on specific conditions. Together, they make analysis more efficient and accurate in Power BI.

Grouping and Aggregating Data

Grouping by *Category* will collect all rows of *Furniture*, all rows of *Clothing*, and all rows of *Electronics* together.

Duplicated the Order Details table

Queries [8]	= Table.Distinct(#"Changed into Fixed decimal", {"Month of Order Date"})		
List of Orders	Month of Order Date	Category	Target
Order Details	Valid 100%	Valid 100%	Valid 100%
Sales target	Error 0%	Error 0%	Error 0%
Order Data	Empty 0%	Empty 0%	Empty 0%
Duplicate order deta...	1 26/04/18	Furniture	10,400.00
	2 26/05/18	Furniture	10,500.00

Count of Order ID

Count
Valid 100%
Error 0%
Empty 0%
1
1
1
1
1
1
1
1
1
1

Average Profit by Category

\$ Average Profit	
Valid	100%
Error	0%
Empty	0%
	20.44
	-19.38
	15.29

Total Amount by Sub-Category

ABC Sub-Category		\$ Total Amount By Sub Category	
	Valid 100%	Valid 100%	
	Error 0%	Error 0%	
	Empty 0%	Empty 0%	
1	Bookcases		25,800.00
2	Phones		11,691.00
3	Trousers		7,109.00
4	T-Shirt		3,205.00
5	Saree		17,831.00
6	Shirt		2,535.00
7	Leggings		851.00
8	Tables		3,967.00
9	Electronic Games		11,942.00
10	Printers		16,822.00
11	Chairs		5,724.00
12	Hankerchief		4,362.00

Data Modeling

Data modeling is the process of defining relationships between tables so that they work together as a single, connected dataset. In Power BI, this is done in the **Model View** using keys (common fields) to link tables.

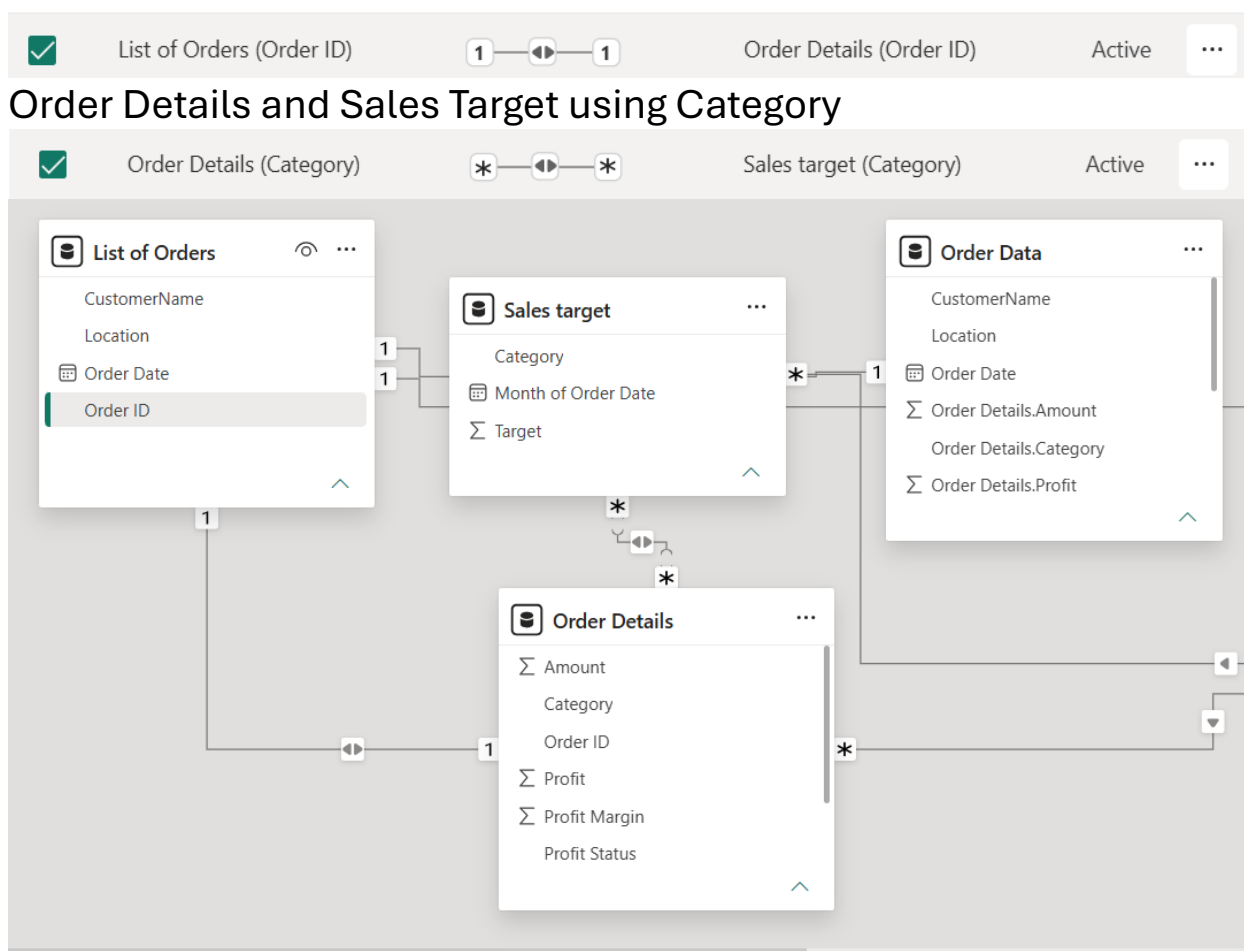
 **Modeling Logic**

Modeling defines relationships between tables in Power BI to enable cross-table analysis:

🔑 **Identify Keys** → Common fields like *Order ID* and *Category*.

🔑 **Establish a relationship between:**

List of Orders and Order Details using Order ID



Conclusion

This project demonstrated how raw business data can be transformed into meaningful insights using Power Query and Power BI. Through systematic steps such as **data cleaning, text formatting, column creation, conditional logic, grouping, and**

merging tables, the datasets were prepared for accurate analysis.

By building a proper **data model** with relationships between *List of Orders*, *Order Details*, and *Sales Target*, the project enabled comparisons of actual sales against targets across categories and time periods. This structured approach ensured that the analysis was reliable, consistent, and easy to interpret.

Overall, the project highlights the importance of **data preparation and modeling** in business intelligence. Clean, well-structured data combined with effective modeling provides a strong foundation for dashboards and reports that support better decision-making.

Thank You
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